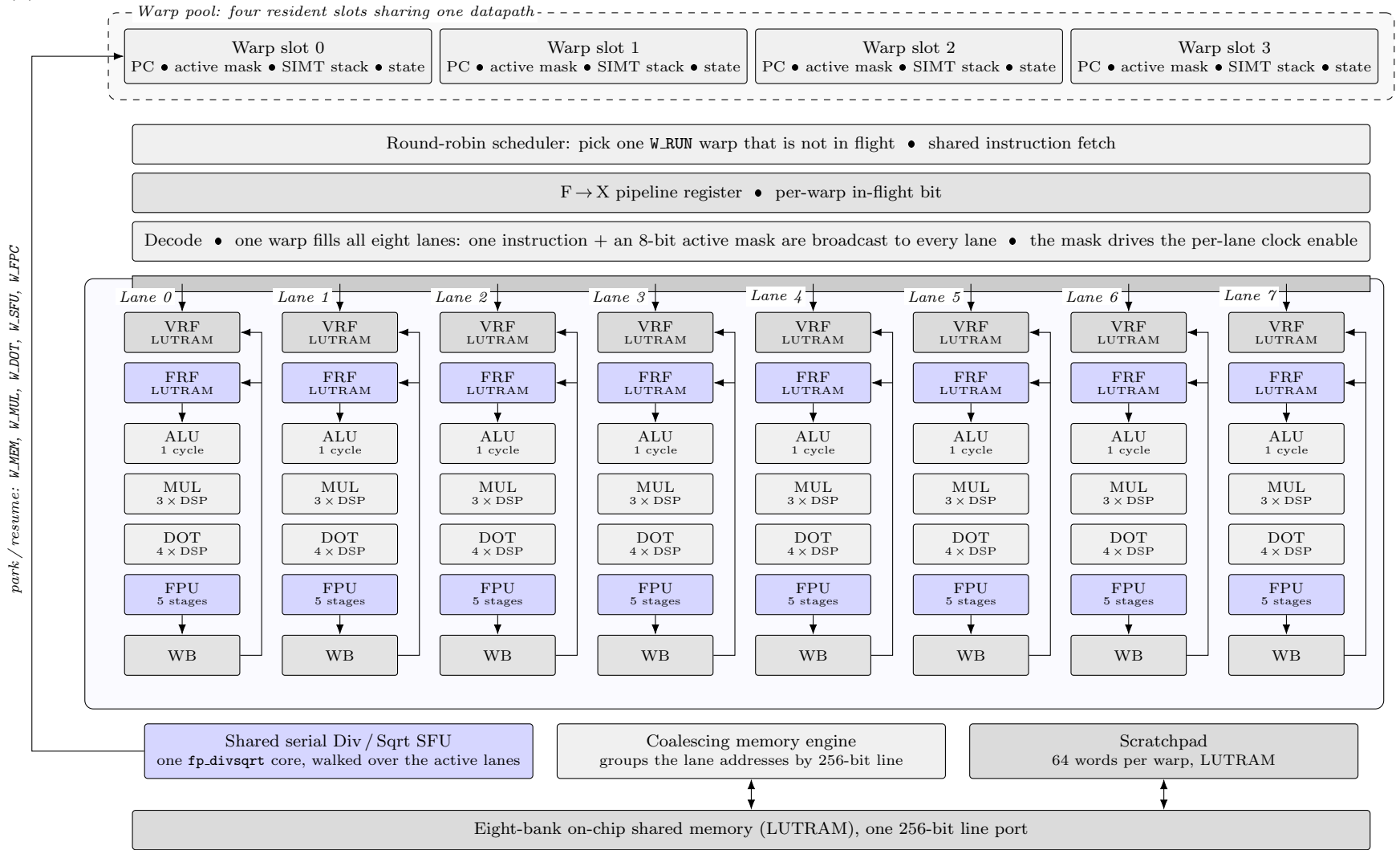


(a) Warp pool and lane array



Per-lane blocks — identical in all eight lanes, replicated by a `generate` loop

VRF integer register bank: distributed RAM, 1 write / 2 async read ports

FRF FP register bank: distributed RAM, 1 write / 3 async read ports (`fs1/fs2/fs3` feed the FMA)

ALU single-cycle integer ALU — the only combinational execute unit

MUL pipelined 32×32 multiplier: three 16×16 DSP48 sub-products + adder tree, 7 cycles

DOT `pdot8` INT8 engine: four signed 9×9 DSP48 products + registered adder tree, 7 cycles

FPU five-stage FP32/FP16 datapath — panel (b)

WB writeback arbiter — steers the result to the integer or the FP bank

Colour key

integer / control logic

floating point

storage (distributed RAM)

Shared, not replicated per lane: the divide/square-root core, the coalescing memory engine, the scratchpad, and the whole fetch/decode/issue front end.

A warp that issues to a multi-cycle unit *parks* on the scoreboard and frees the datapath, so the scheduler keeps issuing the other warps while that unit runs. This is what hides the 7-cycle MUL/DOT, the 5-cycle FPU, and the tens-of-cycles SFU latency.

(b) Inside one lane: the five-stage FP unit

simt.fpu — one instance per lane; FP32 and FP16 share one datapath

